Zhuohan Xie, Ph.D.

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Summary: I am a Research Scientist at MBZUAI working on AI for Finance, financial agents, and trustworthy LLM evaluation. I build benchmarks, shared tasks, and verification systems for financial reasoning, evidence-grounded decision-making, and multilingual and multimodal evaluation. I also lead community evaluation initiatives and mentor early-career researchers from problem formulation through publication and release.

Selected First- and Last-Author Publications

- **[NeurIPS 2026] The Geometry of Forgetting: Temporal Knowledge Drift as an Independent Axis in LLM Representations**
  Rania Elbadry, Ahmed Heakl, Fan Zhang, Dani Bouch, Yuxia Wang, Preslav Nakov, **Zhuohan Xie**
  Last author. Representation-level detection of outdated knowledge; orthogonality to correctness and uncertainty.

- **[ACL 2026 Main, Oral] FinChain: A Symbolic Benchmark for Verifiable Chain-of-Thought Financial Reasoning**
  Zhuohan Xie, Daniil Orel, Rushil Thareja, Dhruv Sahnan, . . . , Salem Lahlou, Veselin Stoyanov, Sophia Ananiadou, Preslav Nakov
  First author. Financial reasoning benchmark; symbolic verification for chain-of-thought evaluation.

- **[ACL 2026 Main, Oral] SAHM: A Benchmark for Arabic Financial and Shari'ah-Compliant Reasoning**
  Rania Elbadry, . . . , Sophia Ananiadou, Veselin Stoyanov, Jimin Huang, Xueqing Peng, Preslav Nakov, **Zhuohan Xie**
  Last author. Arabic financial reasoning and Shari'ah-compliant evaluation.

- **[ACL 2026 Findings] FinCARDS: Card-Based Analyst Reranking for Financial Document Question Answering**
  Yixi Zhou, Fan Zhang, Yu Chen, Haipeng Zhang, Preslav Nakov, **Zhuohan Xie**
  Last author. Financial document QA and analyst-oriented reranking.

- **[ACL 2026 Demo] FinReporting: An Agentic Workflow for Localized Reporting of Cross-Jurisdiction Financial Disclosure**
  Fan Zhang, Mingzi Song, Rania Elbadry, . . . , Preslav Nakov, **Zhuohan Xie**
  Last author. Agentic financial reporting system.

- **[SIGIR 2026 Oral] Multi-Sourced, Multi-Agent Evidence Retrieval for Fact-Checking**
  Shuzhi Gong, Richard O. Sinnott, Jianzhong Qi, Cécile Paris, Preslav Nakov, **Zhuohan Xie**
  Last author. Evidence retrieval and multi-agent fact-checking.

- **[EMNLP 2026 Findings] Right Frame, Wrong Rule: Cultural Cues Expose the Financial Knowledge Gap They Were Meant to Close**
  Rania Elbadry, Ahmed Heakl, Saeed Almheiri, . . . , Preslav Nakov, Yuxia Wang, Fajri Koto, **Zhuohan Xie**
  Last author; accepted, forthcoming. Cultural framing and financial reasoning evaluation.

- **[AACL 2026 Findings] SQLStructEval: Structural Evaluation of LLM Text-to-SQL Generation**
  Yixi Zhou, Fan Zhang, Zhiqiao Guo, Yu Chen, Haipeng Zhang, Preslav Nakov, **Zhuohan Xie**
  Last author; accepted, forthcoming. Structural evaluation of generated SQL queries.

- **[NAACL 2025 Findings] FIRE: Fact-checking with Iterative Retrieval and Verification**
  Zhuohan Xie, Rui Xing, Yuxia Wang, . . . , Iryna Gurevych, Preslav Nakov
  First author. Fact-checking with iterative evidence retrieval and verification.

- **[EMNLP 2023 Findings] DeltaScore: Fine-Grained Story Evaluation with Perturbations**
  Zhuohan Xie, Miao Li, Trevor Cohn, Jey Lau
  First author. Perturbation-based evaluation of generated stories.

- **[INLG 2023] The Next Chapter: A Study of Large Language Models in Storytelling**
  Zhuohan Xie, Trevor Cohn, Jey Han Lau
  First author. LLM storytelling evaluation.

- **[CLEF 2026, LNCS] Overview of FinMMEval 2026: Multilingual and Multimodal Financial Evaluation**
  Zhuohan Xie, Yuyang Dai, Rania Elbadry, . . . , Xue Liu, Preslav Nakov
  First author; forthcoming. Overview of the FinMMEval 2026 lab.

- **[arXiv 2026] FinPersona-Bench: A Benchmark for Longitudinal Psychometric Stability of Autonomous Financial Agents**
  Muhammad Usman Safder, Ayesha Gull, Rania Elbadry, . . . , Preslav Nakov, **Zhuohan Xie**
  Last author. Long-horizon financial-agent evaluation for mandate stability and selective re-grounding.

- **[arXiv 2026] Can LLMs Be CEOs? Benchmarking Strategic Resource Reallocation with Multi-Role Agent Simulation**
  Yuyang Dai, Xueqing Peng, Lingfei Qian, **Zhuohan Xie**
  Last author. Constraint-rich executive decision-making with multi-role agent simulation.

- **[arXiv 2026] Seeing Is Not Deciding: Can Multimodal LLMs Act as Effective CEOs?**
  Yuyang Dai, Xueqing Peng, Yuxia Wang, Preslav Nakov, **Zhuohan Xie**
  Last author. Multimodal executive-decision benchmark separating perception from constrained action.

- **[arXiv 2026] Style Wins, Substance Loses: A Diagnosis of LLM-as-Judge in Idea Generation**
  Fengxian Ji, Yuke Li, Jingpu Yang, . . . , Min Peng, Qianqian Xie, Xiuying Chen, **Zhuohan Xie**
  Last author. Bias diagnosis and mitigation for LLM-based evaluation of scientific ideas.

Selected Additional Publications

- **[ACL 2026 Main, Oral] Is Human-Like Text Liked by Humans? Multilingual Human Detection and Preference Against AI**
  Yuxia Wang, Rui Xing, . . . , **Zhuohan Xie**, . . . , Nizar Habash, Iryna Gurevych, Preslav Nakov
  Co-author. Multilingual AI text detection and human preference.

- **[ACL 2026 Main] Cultural Benchmarking of LLMs in Standard and Dialectal Arabic Dialogues**
  Muhammad Dehan Al Kautsar, Saeed Almheiri, . . . , **Zhuohan Xie**, Junhong Liang, Mohammad Rustom Al Nasar, Preslav Nakov, Fajri Koto
  Co-author. Cultural evaluation across Standard Arabic and regional dialects.

- **[ACL 2026 Findings] Same Claim, Different Judgment: Benchmarking Scenario-Induced Bias in Multilingual Financial Misinformation Detection**
  Zhiwei Liu, . . . , **Zhuohan Xie**, . . . , Alejandro Lopez-Lira, Xue Liu, Jimin Huang, Sophia Ananiadou
  Co-author. Financial misinformation and scenario-induced bias.

- **[ACL 2026 Findings] RealFin: How Well Do LLMs Reason About Finance When Users Leave Things Unsaid?**
  Yuyang Dai, Yan Lin, **Zhuohan Xie**, Yuxia Wang
  Co-author. Implicit-assumption financial reasoning.

- **[SIGIR 2026 Resource] Conv-FinRe: A Conversational and Longitudinal Benchmark for Utility-Grounded Financial Recommendation**
  Yan Wang, Yi Han, Lingfei Qian, . . . , **Zhuohan Xie**, . . . , Jimin Huang, Yankai Chen, Xue Liu, Jian-Yun Nie
  Co-author. Financial recommendation benchmark.

- **[EMNLP 2026 Main] Jais 2: A Family of Arabic-Centric Open Large Language Models**
  Mohamed Anwar, . . . , Yuxia Wang, **Zhuohan Xie**, Ali Mekky, . . . , Michalis Vazirgiannis, Preslav Nakov
  Co-author; accepted, forthcoming. Arabic-centric open large language models.

- **[EMNLP 2026 Main] Aligning Cultural Behaviors in Arabic Language Models via Preferences**
  Junhong Liang, Daniil Orel, Saeed Almheiri, . . . , Abed Alhakim Freihat, **Zhuohan Xie**, Momina Ahsan, . . . , Preslav Nakov, Fajri Koto
  Co-author; accepted, forthcoming. Preference-based alignment of cultural behaviors in Arabic language models.

- **[TACL 2026] Can LLMs Automate Fact-Checking Article Writing?**
  Dhruv Sahnan, . . . , **Zhuohan Xie**, Iryna Gurevych, Elizabeth Churchill, Tanmoy Chakraborty, Preslav Nakov
  Co-author. Fact-checking article writing and LLM-assisted verification.

- **[ACL 2025 Main] KazMMLU: Evaluating Language Models on Kazakh, Russian, and Regional Knowledge of Kazakhstan**
  Mukhammed Togmanov, Nurdaulet Mukhituly, . . . , Akhmed Sakip, **Zhuohan Xie**, Yuxia Wang, . . . , Preslav Nakov, Fajri Koto
  Co-author. Evaluation of Kazakh, Russian, and regional knowledge.

- **[ACL 2025 Findings] Entity Framing and Role Portrayal in the News**
  Tarek Mahmoud, **Zhuohan Xie**, Dimitar Dimitrov, . . . , Giovanni Da San Martino, Jakub Piskorski, Preslav Nakov
  Co-author. Multilingual entity framing and role portrayal in news.

- **[EMNLP 2025 Main] A Head to Predict and a Head to Question: Pre-trained Uncertainty Quantification Heads for Hallucination Detection in LLM Outputs**
  Artem Shelmanov, Ekaterina Fadeeva, Akim Tsvigun, Ivan Tsvigun, **Zhuohan Xie**, . . . , Preslav Nakov, Timothy Baldwin
  Co-author. Pre-trained uncertainty heads for hallucination detection.

- **[COLM 2025] Sherkala-Chat: Building a State-of-the-Art LLM for Kazakh in a Moderately Resourced Setting**
  Fajri Koto, Rituraj Joshi, Nurdaulet Mukhituly, Yuxia Wang, **Zhuohan Xie**, . . .
  Co-author. Open Kazakh LLM with multilingual pre-training, instruction tuning, and safety alignment.

- **[FinNLP 2026] Prediction-Market Seed Capital Recovery from Noise-Dominant Flow**
  Yankai Chen, Bowei He, **Zhuohan Xie**, Zhiwei Liu, Fan Zhang, Xueqing Peng, Yan Wang, Lingfei Qian, Xue Liu
  Co-author; accepted Long Paper; manuscript not yet public. Prediction-market mechanism design and seed-capital recovery.

- **[SemEval 2025] SemEval 2025 Task 10: Multilingual Characterization and Extraction of Narratives from Online News**
  Jakub Piskorski, Tarek Mahmoud, . . . , Nicolas Stefanovitch, **Zhuohan Xie**, Preslav Nakov, Giovanni Da San Martino
  Co-author and task organizer. Shared-task overview of multilingual entity framing and narrative analysis.

- **[arXiv 2026] Herculean: An Agentic Benchmark for Financial Intelligence**
  Xueqing Peng, **Zhuohan Xie**, Yupeng Cao, Haohang Li, Lingfei Qian, . . . , Sophia Ananiadou
  Co-author. Large-scale benchmark for agentic financial intelligence.

Research Program

- **Vision: Reliable Evaluation Infrastructure for High-Stakes LLMs.** I build benchmarks, shared tasks, and verification systems that test whether LLMs can reason with evidence, structured constraints, multilingual inputs, and domain risk rather than simply generate plausible text.
- **Outputs: Finance, Agents, and Verification.** This program spans verifiable financial reasoning (FinChain, SAHM, FinCARDS), multilingual and multimodal evaluation (FinMMEval), agentic decision benchmarks (Herculean, FinPersonaBench, CEO-Bench, C-SuiteBench), evidence systems (FIRE and multi-agent retrieval), and representation-level analysis of outdated knowledge (The Geometry of Forgetting).

Invited Talks and Research Seminars

September 2026	**INSAIT, Sofia**. Towards Trustworthy Financial AI. *Invited research talk*.
24 July 2026	**University of Melbourne NLP Group**. Towards Trustworthy Financial AI: Verifiable Reasoning, Grounded Evidence, and Shared Evaluation. *Invited research seminar*.
24 July 2026	**HKU EMBA Tokyo session**. Guest lecture on AI and financial decision-making. *Invited guest lecture, delivered remotely*.
14 May 2026	**Wuhan University**. Reliable Financial Reasoning: FinChain and SAHM. *Research seminar*.
25 September 2025	**Abu Dhabi Machine Learning Meetup (ADML)**. FinChain and Beyond: Towards Transparent Financial Reasoning in AI. *Invited community talk*.
14 August 2025	**The Fin AI, online**. FIRE: Fact-checking with Iterative Retrieval and Verification. *Research talk*.

Research Leadership and Academic Service

May 2026 – Present	**Chair**, Asia-Pacific Chapter, The Fin AI.
2026	**Area Chair**, FinNLP 2026, a workshop at EMNLP 2026.
2026–2027	**Lab Organizer**, CLEF FinMMEval. Organized the 2026 lab, including participant communication, evaluation workflow, and working-note coordination. The 2027 lab has been accepted and is upcoming.
2025–Present	**Research Mentoring**, Rania Elbadry, Yixi Zhou, Shuzhi Gong, Fan Zhang, Fengxian Ji, and Yuyang Dai; co-developed ACL/SIGIR papers and 2026 benchmark releases.
2025–2027	**Organizer and Committee Member**, SemEval 2025 Task 10 Organizer; SemEval 2027 Task Selection Committee.
2023–Present	**Reviewer**, ACL Rolling Review, NeurIPS, COLM, EMNLP, ALTA, CHOMPS, TFAI, and MisD.

Appointments and Education

May 2026 – Present	**Research Scientist**, MBZUAI, Abu Dhabi, UAE
Jul 2024 – May 2026	**Postdoctoral Associate**, MBZUAI, Abu Dhabi, UAE
Feb 2024 – Jul 2024	**Research Assistant**, MBZUAI, Abu Dhabi, UAE
Feb 2020 – Jul 2024	**Ph.D. in Computing and Information Systems**, University of Melbourne. Thesis: *Automatic Story Generation and Evaluation*.
Jul 2017 – Jun 2019	**Master of Information Technology**, University of Melbourne. Thesis: *Chinese Poetry Automatic Generation*.
Sep 2013 – Jun 2017	**Bachelor of Software Engineering** and **Bachelor of Advertising**, Xiamen University.

Awards and Funding

2026	**Rising Star Program (Chenxing)**, Wuhan University. CNY 300,000 (approx. USD 45,000 equivalent).
2020–2023	**Melbourne Research Scholarship**, University of Melbourne. Full tuition + AUD 30,600/year (approx. USD 22,000/year equivalent).
2018, 2019	**Dean's Honours List**, Melbourne School of Engineering, University of Melbourne.

Teaching and Skills

2019–2023	Academic Tutor, University of Melbourne.
Skills	Python, PyTorch, Hugging Face Transformers, OpenAI API, LangChain, scikit-learn, Bash, LaTeX; English and Mandarin Chinese.